

KARL HILL

SOFTWARE ENGINEERING

TECHNICAL LEADERSHIP | MISSION SOFTWARE | AGILE DELIVERY

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SUMMARY

Staff Aerospace Software Engineer at Jacobs. The work is Python mission software, technical delivery, and mentoring on a team of about 10. Previously Lead Software Engineer at SSAI supporting NASA Goddard, 2017 to 2025.

PROFESSIONAL EXPERIENCE

Staff Aerospace Software Engineer

SEPT 2025 — PRESENT

Jacobs — National Security · Chantilly, VA

- Develop Python-based mission software, shared interfaces, messaging integrations, and service orchestration across multiple deployment environments.
- Guide technical delivery for a team of about 10 engineers across approximately 20 repositories; sequence work, coordinate dependencies, and drive integration and release readiness.
- Build shared CI/CD, automated testing, security checks, repository standards, dependency management, and release automation.
- Onboarded and coached approximately six engineers through code review, technical feedback, and structured development guidance.
- Lead Agile planning and execution, working with program stakeholders and partner teams to turn mission priorities into actionable engineering work.

Lead Software Engineer

DEC 2017 — SEPT 2025

SSAI / NASA Goddard Space Flight Center · Greenbelt, MD

- Led the design and development of an AWS-based flood-mapping system, automating processing and distribution of satellite-derived flood products.
- Modernized LAADS DAAC web systems, including Find Data search and ordering and near-real-time access, with GitLab CI/CD and Kubernetes delivery.
- Modernized NASA Earth Observatory's web platform, improving performance, user experience, and search discoverability for an audience of approximately 1.5 million monthly visitors during that work.
- Automated a content-registry workflow to reduce manual dataset-registration work and improve researcher access.
- Built a Ceph-based file and metadata platform for virtual directory mapping and discovery of large scientific datasets.

DETAILS

Washington, DC

(202) 599-1442

karlhillx@gmail.com

LINKS

LinkedIn

linkedin.com/in/khill

GitHub

github.com/karlhillx

Website

karlhill.com

CORE COMPETENCIES

- Software Engineering
- Technical Direction & Leadership
- Distributed Systems & Integration
- CI/CD & Developer Tooling
- Engineer Development & Coaching
- Agile Delivery Leadership
- Integration & Release Readiness

PROFESSIONAL EXPERIENCE

Senior Software Engineer

JAN 2016 — DEC 2017

InformedDNA · Washington, DC

- Designed and delivered a Laravel-based case-management platform connecting counseling workflows, documentation, and billing.
- Developed CRM enhancements for customer lifecycle workflows and reporting.
- Improved platform maintenance and security processes in partnership with operations.

Senior Software Engineer

JUN 2012 — MAR 2015

Ticomix, Inc. · Washington, DC

- Delivered SugarCRM solutions for more than 20 clients, including the Virginia Department of Transportation and Kastle Systems.
- Improved backlog management and delivery practices across client projects.

Earlier Engineering Experience

1997 — 2012

Sabre Corporation · Dante Inc. · Visitar Inc. · Verizon Business

- Held software engineering and principal-level roles across travel, enterprise CRM, telecommunications, and managed security.
- Built Java and SQL Server services for Finium, a multi-tenant managed-security platform, and contributed to shared testing and code-quality practices.

CORE COMPETENCIES

Software Engineering · Technical Direction & Leadership · Distributed Systems & Integration · CI/CD & Developer Tooling · Engineer Development & Coaching · Agile Delivery Leadership · Integration & Release Readiness

EDUCATION

Computer Science coursework, University of Maryland, Baltimore County**Associate of Arts, General Studies — Computer Science Curriculum**, Howard Community College**Project Management studies**, Rutgers University

CERTIFICATIONS

SAFe Agilist, Scaled Agile (in progress)**Professional Scrum Master II**, Scrum.org**Professional Scrum Product Owner I**, Scrum.org

DEVELOPER TOOLING

bb-run — Run Bitbucket Pipelines locally.**testrisk** — Rank the highest-value Python test gaps from coverage, AST, and git.

TECHNICAL EXPERTISE

Languages: Python, TypeScript, Java, PHP, Perl, Bash**Cloud & Containers:** AWS, Docker, Kubernetes, Helm**CI/CD & Developer Tooling:** Bitbucket Pipelines, GitLab CI, GitHub Actions, release automation**Python Tooling:** uv, Ruff, pytest, mypy, pre-commit**Integration:** REST/OpenAPI, RabbitMQ, ActiveMQ, distributed systems**Data:** PostgreSQL, MySQL, SQL Server, Elasticsearch/OpenSearch, Redis